

EXECUTIVE EVALUATION REPORT: HOUSING PRICE PREDICTION

Prepared by: Janvi Singh

Date: June 2026 | Project: Week 1 Predictive Analytics Internship

1. Objective and Problem Alignment

The core objective of this initiative was to eliminate traditional pricing guesswork in real estate transactions by engineering a data-driven regression architecture. Using a real-world housing dataset comprising structural, amenity-based, and geographical variables, we constructed a robust machine learning workflow to accurately project property valuation and isolate key macroeconomic value drivers.

2. Model Performance Evaluation

We benchmarked a traditional Baseline Linear Regression model against an ensemble Random Forest Regressor. The dataset was split into an 80% training set and a 20% validation set to preserve validation integrity.

Contrarian to initial expectations, the Linear Regression model outperformed the Random Forest Regressor across all evaluation metrics, capturing 65.29% of the pricing variance compared to the Random Forest's 61.19%. It achieved a lower Mean Absolute Error (MAE) of 970,043.40.

Machine Learning Model	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R ² Score (Variance Captured)
Linear Regression	970,043.40	1,324,506.96	0.6529
Random Forest Regressor	1,021,546.04	1,400,565.97	0.6119

Core Model Finding: The superior performance of the linear baseline indicates that the structural and amenity assets within this dataset scale in a highly linear fashion with property value. Complex ensemble models like Random Forest over-compensated for non-existent non-linear complexities, leading to a slight drop in generalization accuracy on the test set.

3. Key Market Insights Derived from Data

- Primary Value Drivers:** Statistical feature correlation directly identifies area holds the strongest positive relationship with price at 0.54, followed immediately by the number of bathrooms (0.52) and the integration of air conditioning (0.45).
- The Valuation Surprise:** Interestingly, secondary structural assets like a guest house (0.26) or a functional basement (0.19) show a lower direct pricing correlation than a

home's vertical layout, such as the number of stories (0.42). Furthermore, a completely unfurnished status introduces a significant negative pricing anchor of -0.28.

4. Strategic Business Recommendations

Based on the predictive weights of our optimal model, we recommend a two-pronged operational strategy for maximizing real estate investment margins:

1. **High-ROI Asset Renovation:** Before listing properties on the market, priority should be given to scaling vertical space (stories) and adding modern climate control (air conditioning) rather than expanding raw bedroom counts (bedrooms correlation sits lower at 0.37).
2. **Mitigating Negative Value Premium:** Real estate teams should aggressively avoid or stage homes sitting in an "unfurnished" status. Because unfurnished status acts as a strong downward pressure on price (-0.28), deploying basic staging furniture can neutralize this negative impact and yield a massive return on investment relative to the listing price.